KISANcare Soil Intelligence Report

Multi-Year Dynamic Nutrient Tracking & Calibrated Lab Ingestion V1

Report ID: SIR_PLOT_MH_LAT_...
Generated: 2026-10-23 18:51 UTC

Plot Identifier: PLOT_MH_LAT_****81 **Region:** Latur District / Marathwada Region, Maharashtra **Baseline Test:** 2023-06-15
Farm Account: FARM_LA_****01 **Privacy Scope:** Masked / Confidential **Testing Lab:** District Soil Testing La..

STATUS: INSUFFICIENT DATA / PREDICTION GATED
Physical laboratory baseline and farm ledger displayed below. Empirical future prediction is abstained.

1. Laboratory Baseline Soil Measurements *(PROVENANCE: MEASURED)*

Lab: District Soil Testing Laboratory, L | Date: 2023-06-15 | Certificate: STL-2023-8871

[ACCREDITATION: UNVERIFIED - UNVERIFIED SELF-REPORTED CREDENTIALS (Verification Pending)]

Soil Parameter	Verbatim Reading	Extraction Protocol	Normalized Elemental	Fertility Classification
Available Nitrogen (N)	245.0 kg/ha (ELEMENT	Alkaline Permanganate (S	**245.0 kg/ha elementa**	Low (< 280 kg/ha)
Available Phosphorus (P)	18.5 kg/ha (ELEMENTA	Olsen 0.5M NaHCO3 (for n	**18.5 kg/ha elemental**	Medium (10 - 24.4 kg
Available Potassium (K)	320.0 kg/ha (ELEMENT	Neutral 1N Ammonium Acet	**320.0 kg/ha elementa**	High (> 279 kg eleme
Soil Reaction (pH)	7.80 pH	Potentiometric 1:2.5 sus	**7.80 pH scale**	Moderately Alkaline
Electrical Conductivity	0.35 dS/m	Conductivity bridge 1:2.	**0.35 dS/m**	Normal (< 1.0 dS/m)
Organic Carbon (OC)	0.52 %	Walkley-Black wet oxidat	**0.52 % by weight**	Medium (0.5 - 0.75%)

2. Chronological Multi-Year Farm Management History *(PROVENANCE: FARMER_RECORDED)*

Date	Event Category	Agronomic Activity	Delta N	Delta P	Delta K
2023-06-20	**FERTILIZER_ADDIT**	Nutrient Addition: Di-Ammonium Phosphate (DA	**+18.0 kg N**	**+20.1 kg P**	**+0.0 kg K**
2023-10-15	**CROP_DEPLETION**	Crop Harvest Removal: Soybean (2.2 t/ha)	-64.3 kg N	-13.2 kg P	-48.4 kg K
2024-03-20	**CROP_DEPLETION**	Crop Harvest Removal: Wheat (3.2 t/ha)	-78.4 kg N	-12.2 kg P	-64.0 kg K

3. Cumulative Agronomic Mass-Balance Ledger *(BOOKKEEPING ONLY)*

Nitrogen (N)
Additions: +18.0 kg/ha
Removals: -142.8 kg/ha
Net Flux: -124.8 (+/- 15.3) kg/ha

Phosphorus (P)
Additions: +20.1 kg/ha
Removals: -25.4 kg/ha
Net Flux: -5.3 (+/- 3.4) kg/ha

Potassium (K)
Additions: +0.0 kg/ha
Removals: -112.4 kg/ha
Net Flux: -112.4 (+/- 16.1) kg/ha

CRITICAL NOTE: Net mass-balance represents field surface flux. It DOES NOT equal extractable soil available NPK.

4. Uncertainty Decomposition & Limitations *(STATUS: UNCALIBRATED_HEURISTIC_BOUND)*

Propagated Physical Uncertainty Bounds (Quadrate Combination):
Nitrogen: +/- 27.4 kg N/ha | Phosphorus: +/- 6.4 kg P/ha | Potassium: +/- 34.4 kg K/ha

5. Operational Prediction Gate & Safe Abstention Rationale

Why is an empirical future NPK prediction not provided in this report?

Operational prediction is BLOCKED: REJECTED: No formal ValidationEvidenceBundle provided. Loose boolean flags or manual assertions cannot grant VALIDATED_FOR_SCOPE. Operational prediction remains strictly BLOCKED.

6. Practical Next Steps & Agronomic Recommendations

* Maintain routine annual crop and fertilizer application logging.

Recommended Soil Retesting Depth: 0 - 15 cm (standard topsoil plow layer)
IMPORTANT: For site-specific fertilizer scheduling and crop guidance, always consult a certified agronomist at your